Abdul Mohaimen Al Radi

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Summary

I'm a second year PhD student at the **University of Central Florida**, advised by **Dr. Yu Tian** and **Dr. Mubarak Shah**. I work on **long-horizon video generation** and **video diffusion models**, and I also study AI agents in medicine.

Education

University of Central Florida — Orlando, FL
Ph.D. in Computer Vision — *Fall 2025 – 2029 (Expected)*

University of Dhaka — Dhaka, Bangladesh
B.Sc. in Computer Science and Engineering (Top 10% of class, CGPA: 3.72/4.0) — *2024*

Selected Publications *(full list)*

Blind Image Deblurring with FFT-ReLU Sparsity Prior
Abdul Mohaimen Al Radi, P. S. Majumder, M. M. Khan. *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2025.*

Aero-World: Action-Conditioned Aerial Video Generation from Inertial Controls
Abdul Mohaimen Al Radi, K. Li, Y. Shang, M. Shah, Y. Tian. *(Under Review)*

Agentic Large-Language-Model Systems in Medicine: A Systematic Review and Taxonomy
Abdul Mohaimen Al Radi, X. Cao, F. Yu, Y. Liu, et al. *(Under Review)*

SloMoDeblur: A Large-Scale Smartphone Image Deblurring Dataset
S. M. Mahmud, M. M. H. Noki, P. S. Majumder, **Abdul Mohaimen Al Radi**, et al. *Data-centric Machine Learning Research (DMLR) 2026.*

From Attention to Frequency: Integration of Vision Transformer and FFT-ReLU for Enhanced Image Deblurring
S. M. Mahmud, M. M. H. Noki, P. S. Majumder, **Abdul Mohaimen Al Radi**, et al. *International Conference on Agents and Artificial Intelligence (ICAART 2026)*

CREST: Curvature-Regulated Event-Centric Sampling for Efficient Long-Video Understanding
M. A. Miraj, **Abdul Mohaimen Al Radi**, S. I. Rayhan, et al. *(Under Review)*

Research & Engineering Experience

AIM Lab, University of Central Florida — Orlando, FL
Graduate Research Assistant (Advisors: Dr. Yu Tian, Dr. Mubarak Shah) — *Fall 2025 - Present*

- Developed GeoCov, a fixed-budget memory controller optimizing long-horizon video generation by purging redundant observations via camera geometry and visual similarity metrics, achieving a 35.1% reduction in Fréchet Video Distance (FVD) while compressing archives from 5,397 to 32 frames.
- Engineered anomaly prediction pipelines utilizing video inversion and frozen generation models to encode motion, effectively scoring future latent states with a lightweight prediction head on UCF-Crime and XD-Violence datasets.
- Co-leading the development of a comprehensive medical AI agent skill benchmark suite, standardizing evaluation protocols for specialized clinical reasoning architectures.

Freelance / Contract — Remote
AI Engineer — *Jan 2024 - Jan 2025*

- Evaluated and rigorously debugged LLM-generated code solutions, directly contributing to the benchmarking and alignment of large language models for algorithmic problem-solving tasks.

The Cognitive Agents and Interaction Lab, University of Dhaka — Dhaka, Bangladesh
Research Assistant — *Spring 2023 - Spring 2025*

- Engineered fast-inference image restoration pipelines integrating Fourier domain analysis with deep learning architectures to resolve small-data constraints.
- Curated SloMoDeblur, a large-scale dataset comprising over 42,000 image pairs, establishing a rigorous benchmark for smartphone-based image deblurring.

Technical Projects

Equi-Agent: Multi-Agent Ophthalmic Diagnostic System PyTorch, RETFound
Lead Developer *2025 - 2026*

- Scripted multi-agent consultation loops and integrated fine-tuned RETFound vision models with visual field archetypes to execute robust clinical equity audits in diagnostic systems.

Sparse View 3D Reconstruction & Gaussian Splatting PyTorch, CUDA
Computer Vision Researcher *2024*

- Engineered benchmarking pipelines to evaluate 3D Gaussian Splatting rendering performance across highly constrained, sparse image sets.
- Analyzed multi-object interaction geometries to identify failure modes in current diffusion-based single-image 3D reconstruction, establishing baseline limitations for segmentation-based isolation.

Awards & Competitive Programming

- **Algorithmic Problem Solving:** Solved **871+** algorithmic and data structure problems across Codeforces, UVa, LightOJ, SPOJ, and AtCoder.
- **Collegiate Competitions:** Top 8% in ICPC Dhaka Regional Prelim 2020; Top 10% at AUST ICPC 2022; Top 50 Finalist in Code Samurai 2022.

Technical Skills

- **Languages:** C/C++, Python, Java, JavaScript, Kotlin, Dart, MATLAB, SQL
  ML/DL Frameworks: PyTorch, CUDA, Keras, TensorFlow, Scikit-Learn, OpenCV, MONAI, Hugging Face Diffusers
  Tools & Systems: Git, SLURM, Docker, FastAPI, ReactJS, NodeJS, Linux/Zsh